Teamwork Agreement Template

For the MathWorks Classroom Challenge Projects

Project: _Quarter-Car Suspension Modeling and Simulation with Simscape Multibody__

Team Name: Quarter car team 2

Instructor: Kevin Graham

Program/Class: __EPP 2026 cohort___

Members and Roles

1. __Sean Chen_- Modeling Lead
2. Vicente Lopez - Analysis/Validation Lead
3. Jacqueline Gutierrez Salas - Documentation & Visualization Lead
4. Tanmayee Chalamalasetti - PM & Quality Assurance & Reproducibility Lead

1) Purpose & Scope

We agree to collaborate to deliver a high-quality, ethical, and reproducible project. Our work will follow the project guidance and produce the specified artifacts (e.g. plots, code, and visualizations).

2) Roles & Responsibilities

- Project Manager (PM): Schedules meetings; maintains timeline; ensures tasks are tracked and disagreements are resolved.
- Modeling Lead: Owns core MATLAB models/simulations
- Analysis/Validation Lead: Performs hand calculations and cross-checks simulation outputs
- Documentation & Visualization Lead: Compiles report (with figures/tables), prepares slides (if needed), and builds visual artifacts (e.g. animations or plots)
- Quality Assurance & Reproducibility Lead: Enforces coherent style across artifacts, ensures work is not overwritten across different versions, ensures data quality, tests/re-runs codes to ensure robustness and reproducibility

3) Communication Norms

- Primary channel: __Discord__ (e.g. Slack)

- Secondary channel: __E-mail__ (e.g. E-mail for summaries/decisions)

- Response time: ____24 hours___________ on weekdays and ________24 hours_____________ on weekends (e.g. within 24 hours on weekdays; within 48 hours on weekends)

- Meeting cadence: _________Monday, Wednesday, Friday (12PM)__________ (e.g. weekly working meeting

- Decision records: _____PM logs decisions in a seperate document uploaded to github_____

4) Collaboration & Tools

- Version control: ___Github__________________________________

- Data management: All datasets, parameters, and assumptions documented in

  _______README.md______________________

File naming & style: ____________Snake_case____________________

- Backups: _PM tags old versions according to numbered system_________

5) Quality Standards (Definition of 'Done')

This project will be done when all the required tests work, it provides realistic results, and the code runs with no errors. At least one teammate must also test the project and confirm that everything works correctly.

e.g. Requirements meet project description, code is reproducible (runs from the README.md file, all specified figures regenerate, at least one teammate has tested edge cases, all assumptions, units, and references are documented in report)

6) Timeline & Milestones

- Week 1:

Finish the Teamwork Agreement. Create a project framework. Divide up tasks among teammates. Review MATLAB and other background information regarding the project

- Week 2:
- Hand calculations
- Initial plots and visuals
- Finish tasks 1-3
- Break Down the Problem: Jacqueline
- Task 1: Sean
- Task 2: Vicente
- Task 3: Tanmayee

- Check in with Mathworks Instructors

  Week 3:
  Continue work on simulation
- Double check our work
- Finish tasks 4-6
- Task 4: Sean and Tanmayee
- Task 5: Vicente
- Task 6: Jacqueline
- Check in with Mathworks instructors

- Week 4:
- Create report
- Create slides and presentation

(e.g.
Week 1: Problem framing, literature/background review, dataset/parameters,
Week 2: First pass models (hand calculations + baseline MATLAB scripts); initial plots/visuals.
Week 3: Validation and quality assurance
Week 4: Report, slides, presentation rehearsal, repository cleanup)

7) Decision-Making & Conflict Resolution

- Default: Consensus after group meetings

- If blocked >48 hrs: Have a group meeting and host a vote, majority wins

- Technical disagreements: Evidence-based comparison before vote and pros and cons for all sides

- Escalation: If unresolved, consult instructor/mentor with a concise memo.

8) Workload, Accountability & Attendance

- Task board: ___To-Do / Doing / Review / Done with owners and due dates documented in separate google doc_____ (e.g. To-Do / Doing / Review / Done with owners and due dates documented in README.md)

- Accountability: _____If there is a missed deadline, notify the group as soon as possible. Create a plan to finish within 24 hours.__________ (e.g. Missed deadline → recovery plan within 24 hrs; repeated misses trigger role redistribution.)

- Attendance: ___If you'll miss a meeting, give an hour notice. Share updates asynchronously through discord.___ (e.g. If you'll miss a meeting, give 24-hour notice and share updates asynchronously.)

10) Accessibility & Inclusion

We commit to respectful collaboration, equitable turn-taking, and flexibility for documented accessibility needs.

11) Deliverables Checklist

- Code repo with clear structure
- Create a README
- Hand calculations linked to results for validation
- Figures and visualizations
- Create a baseline multibody model
- Create a road test with 3-5 cases
- Log signals and define objective metrics
- Create an automated test runner
- Tune suspension parameters
- Validate robustness with a simple variation
- Write the final report with specified sections
- Create a final slides
- Create a presentation

(e.g. Code repository with clear structure and README, hand calculations linked to modeled results for cross-validation, specific figures & visualizations, final report with specified sections)

12) Agreement & Signatures

Name & Signature: _Sean Chen_____________________________ Date: __7/10/26___

Name & Signature: Vicente Lopez Date: 7/10/26

Name & Signature: Tanmayee Chalamalasetti Date: 7/10/2026

Name & Signature: Jacqueline Gutierrez Salas Date: 7/10/2026